BANGALORE UNIVERSITY

Master of Business Administration (MBA) - II Semester

Technology for Management (Course Code: 2.1)

PROJECT REPORT ON

REGIONAL HEALTHCARE FACILITIES AND ICU BED PORTAL

Design and Implementation using MySQL (RDBMS)

Submitted by

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Link: https://github.com/evanjalinejames/icu-bed-portal-mysql

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1. INTRODUCTION AND PROBLEM STATEMENT

1.1 Background

Karnataka has a large network of government hospitals, medical college hospitals, trusts and private multi-speciality hospitals, concentrated mainly in Bengaluru and a few district headquarters. Critical-care beds, especially ICU beds with ventilators, neonatal ICU (NICU) and paediatric ICU (PICU) beds, are limited and unevenly distributed. When a patient needs an ICU bed, the family or the referring doctor usually telephones hospital after hospital, and every call costs minutes that can decide the outcome of a critical patient.

This project designs a Regional Healthcare Facilities and ICU Bed Portal (RHIBP): one web-based system on which hospitals update the status of their critical-care beds, families and referring doctors register a bed request, and the control room allocates the nearest suitable bed according to clinical priority. The portal is built on a relational database (MySQL) and shows how an information system turns day-to-day transactions into managerial information and decision support. The report follows the six modules of the Technology for Management syllabus.

1.2 Problem Statement

There is no single, real-time view of free ICU beds across hospitals. Information is held in registers, phone calls and messaging groups, which quickly become outdated. As a result, families lose time, some hospitals are overloaded while others have unused beds, requests are served by influence or luck rather than by clinical need, and the Health Department has no reliable data on occupancy, waiting time or cost for planning.

1.3 Objectives

- To maintain a live register of facilities, ICU beds and bed status across regions.
- To record bed requests, rank them by clinical severity, and allocate beds with a complete audit trail.
- To design a normalised (3NF) relational database that removes redundancy and protects data integrity.
- To provide MIS and DSS reports on regional availability, occupancy, waiting requests, length of stay and billing using SQL.
- To analyse the e-governance, adoption, security, ethical and environmental aspects of the portal.

1.4 Scope and Methodology

The scope covers bed management for critical-care wards of eight sample facilities in five regions: facility master data, ward types and rates, bed status, patient requests, admissions and payments. Clinical records, laboratory systems and ambulance dispatch are outside the scope. The work followed analysis, design, MySQL implementation and query-based reporting. All names, facilities and figures are fictitious and prepared only for this academic project. The reference date and time for all outputs is 5 November 2026, 10:00.

1.5 Syllabus Coverage

Syllabus Module	Project Deliverable	Chapter
1. IS Foundations	System classification (TPS, MIS, DSS, KMS, AI)	2
2. MIS and RDBMS	System view, ER design, 3NF schema	3
3. SAD and BPR	SDLC selection, BPR, Context diagram, Level-1 DFD	4
4. E-Commerce and Adoption	E-payment, security architecture, adoption model	5
5. IT Ethics and Security	Privacy, cybercrime, ergonomic and environmental audit	6
6. Application Software	Working MySQL scripts, DSS queries, executive report	7

Table 1: Mapping of syllabus modules to the project

2. MODULE 1: INFORMATION SYSTEM FOUNDATIONS AND CLASSIFICATION

2.1 Information System Components

An information system is an organised combination of hardware, software, data, people and procedures that collects, processes, stores and distributes information to support operations and decision making. In the RHIBP the hardware is hospital and control-room computers and a database server; the software is MySQL and the portal; the data are facility, bed, request, admission and payment records; the people are bed managers, operators, families and officials; and the procedures are the triage, allocation and billing rules.

2.2 Classification of the System

A health-care portal is not a single type of information system. It combines systems that serve different management levels. Table 2 classifies the RHIBP components.

System Type	Management Level	Function in RHIBP
TPS (Transaction Processing System)	Operational	Records bed requests, bed allocation, admission, discharge and payment; each transaction updates the database at once.
MIS (Management Information System)	Middle management	Region-wise bed availability, pending-request queue and facility occupancy reports for district officers and the control room.
DSS (Decision Support System)	Top management	Ward-type shortage analysis, ownership-wise load and billing analysis for capacity and funding decisions.
KMS (Knowledge Management System)	All levels	Triage guidelines, standard operating procedures and lessons from surge periods stored for reuse.
AI application (future)	Strategic	Forecasting ICU demand, recommending the best bed by distance and severity, chatbot for families.

Table 2: Classification of the system

Primary classification: The portal is mainly a TPS at the hospital and control-room level, which feeds an MIS and a DSS. The SQL queries in Chapter 7 act as the DSS layer because they combine data from many tables to answer management questions.

2.3 Enterprise Framework

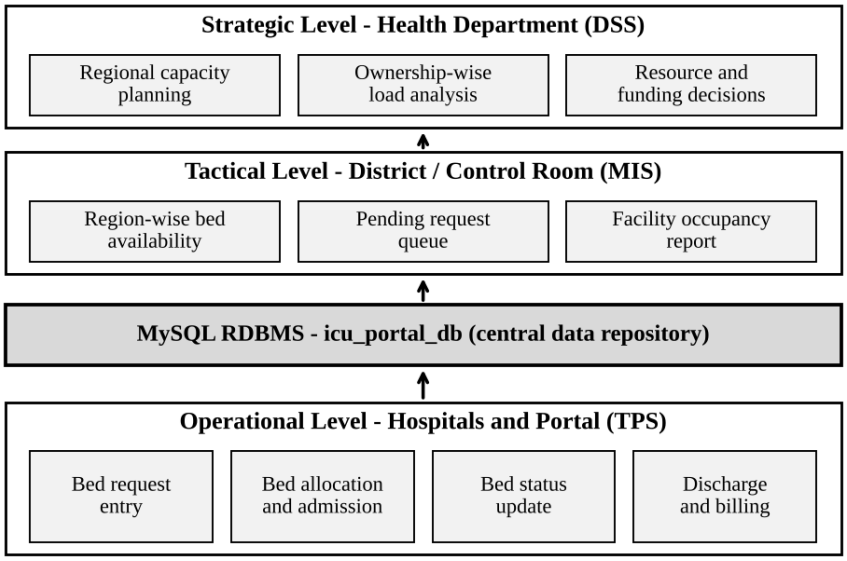


Figure 1: Enterprise framework of the RHIBP

3. MODULE 2: SYSTEM VIEW OF BUSINESS, MIS AND RDBMS DESIGN

3.1 System View of the Health-Care Bed Network

Element	ICU Bed Portal
Inputs	Bed status from hospitals, patient details, ward type needed, triage severity, rates, payments
Processing	Registration of request, search for suitable bed, ranking by priority, allocation, admission, discharge, billing
Outputs	Allocation notice, bed availability dashboard, bills, MIS and DSS reports
Environment	Patients, hospitals, government, insurers, regulators; feedback is waiting time and occupancy data

Table 3(a): System view of the business

Why an RDBMS. A relational database enforces integrity through primary keys, foreign keys and constraints, supports many hospitals updating at the same time, and lets management ask new questions through SQL without changing the application.

3.2 Entity Relationship Diagram

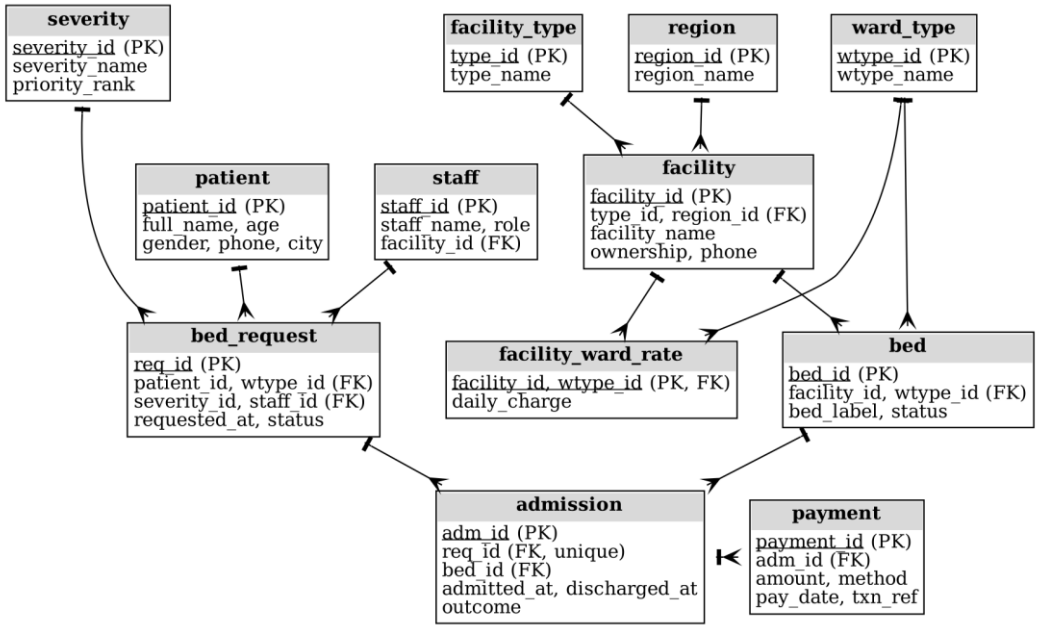


Figure 2: ER diagram of icu_portal_db (crow's-foot notation)

3.3 Entities and Relationships

Table	Purpose	Primary Key
region, facility_type	Lookup tables for district and type of hospital	region_id, type_id
facility	Hospital with type, region and ownership	facility_id
ward_type, facility_ward_rate	Kinds of critical-care ward; daily charge of each ward in each facility	wtype_id; (facility_id, wtype_id)
bed	Individual bed with ward type and live status	bed_id
patient, severity	Patient identity; triage level with priority rank	patient_id, severity_id
staff	Control-room and hospital users	staff_id
bed_request	Request for a bed with severity and status	req_id
admission, payment	Actual stay in a bed; payments made for it	adm_id, payment_id

Table 3: Entities of the database

3.4 Normalisation to Third Normal Form

Unnormalised design. A paper or spreadsheet register records one row per request with hospital name, district, hospital type, ward name, daily charge, patient details, staff name and payments repeated in every row.

1NF and 2NF. Each column holds one value; repeating payments move to the payment table. Partial dependencies are removed. The daily charge depends on both the facility and the ward type together, so it is stored in facility_ward_rate with the composite key (facility_id, wtype_id) and not in bed or ward_type. Patient details depend only on patient_id, so they are stored once in patient.

3NF. Transitive dependencies are removed. Facility determines region and hospital type, so region_name and type_name are moved to region and facility_type. Severity name and priority rank depend on severity_id, not on each request. After this step every non-key attribute depends on the key, the whole key and nothing but the key.

Integrity rules. Every table has a primary key; foreign keys enforce referential integrity (for example admission.bed_id must exist in bed); ENUM lists and CHECK constraints control bed status, payment method, age and amounts.

4. MODULE 3: SYSTEMS ANALYSIS AND DESIGN (SAD) AND BPR

4.1 SDLC Model Selection

Model	Strength	Limitation for this project
Waterfall	Simple, well documented	Rigid; hospital needs and policies change during rollout
Prototyping	Early feedback from bed managers	Needs disciplined documentation
Spiral	Strong risk analysis	Costly for a state-level pilot
Agile / Iterative	Short cycles, phased rollout	Needs active involvement of hospitals

Table 4: Comparison of SDLC models

Selected model: Iterative prototyping with phased rollout. Because the system supports life-critical decisions, it is first built as a prototype, tested in two or three districts, reviewed by hospital bed managers and the control room, and then extended region by region. Phases followed: (1) planning and feasibility, (2) analysis, (3) design, (4) implementation, (5) testing, and (6) maintenance.

Feasibility. The project is technically feasible (open-source MySQL, web access), economically feasible (no licence cost, savings from faster allocation and better planning) and operationally feasible with training and clear rules for updating bed status (Chapter 5).

4.2 Business Process Re-engineering (BPR)

Process	As-Is (Manual)	To-Be (RHIBP)
Finding a bed	Family phones many hospitals; 1 to 3 hours	One search across all facilities; seconds
Prioritisation	Influence, first come first served	Severity rank, then waiting time
Bed status	Registers, calls, messaging groups; outdated	Bed manager updates status; live view
Admission record	Separate paper files	Allocation, admission and bed status updated in one transaction
Billing	Manual, disputes on days and rates	View calculates days and rate; payments tracked
Reporting	Monthly compilation by hand	On-demand MIS and DSS queries

Table 5: BPR - as-is and to-be processes

4.3 Context Diagram (Level 0)

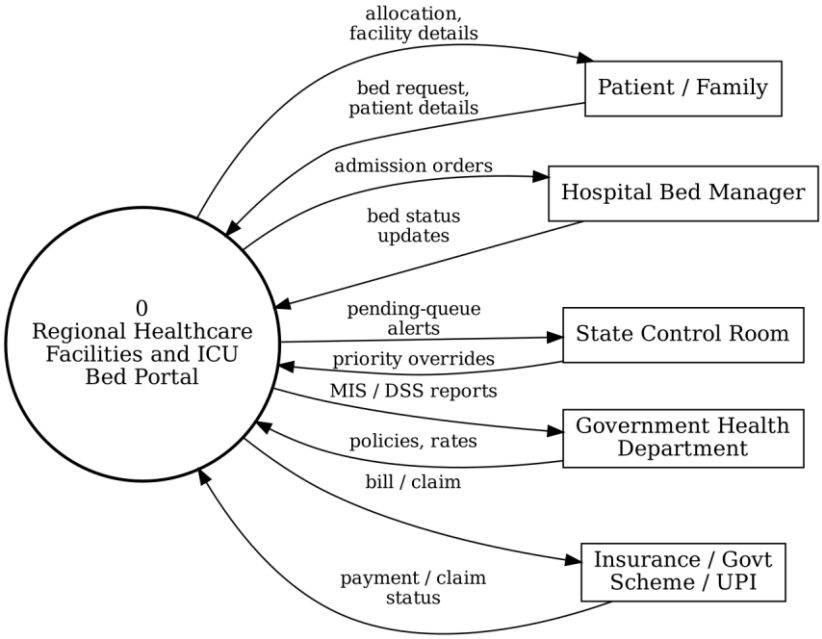


Figure 3: Context diagram of the RHIBP

The context diagram shows the whole system as one process and its data exchange with five external entities: the Patient or Family, the Hospital Bed Manager, the State Control Room, the Government Health Department, and the Insurance, Government Scheme or UPI payment network.

4.4 Level-1 Data Flow Diagram

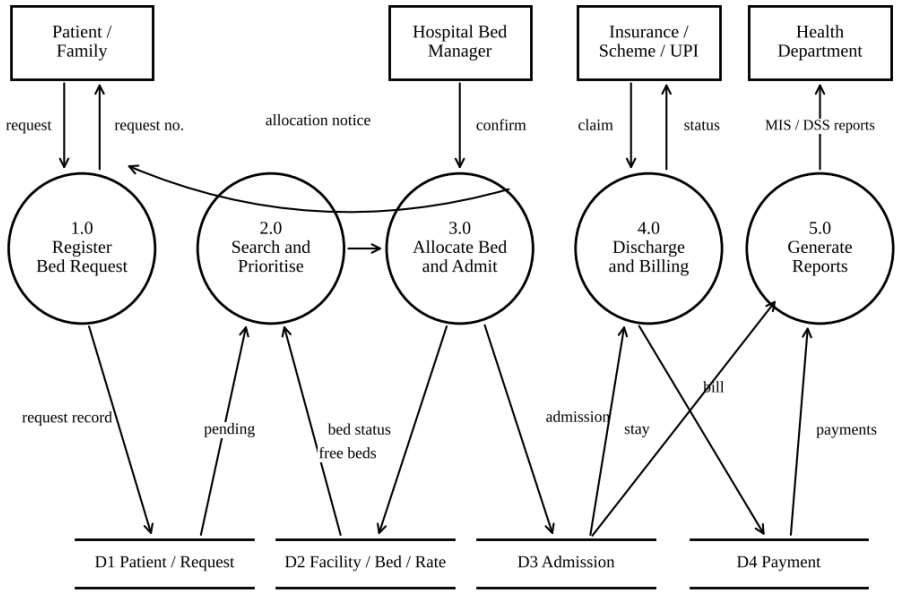


Figure 4: Level-1 DFD (D1 to D4 are data stores)

The Level-1 DFD splits the system into five processes: Register Bed Request, Search and Prioritise, Allocate Bed and Admit, Discharge and Billing, and Generate Reports. Each data store corresponds to one or more tables of the database.

5. MODULE 4: E-COMMERCE, E-PAYMENT AND TECHNOLOGY ADOPTION

5.1 E-Governance and E-Commerce Model

The portal is mainly a G2C (government to citizen) e-governance service, because families use it to obtain a critical-care bed, and a G2B service for private hospitals and trusts that publish their bed status. Paid private care adds an e-commerce element: a family can see the daily charge before accepting a bed and can pay online. In the sample data, 4 of the 8 facilities are private or trust-run and together collected Rs. 1,71,000 against Rs. 46,900 collected by government facilities (see Query 4).

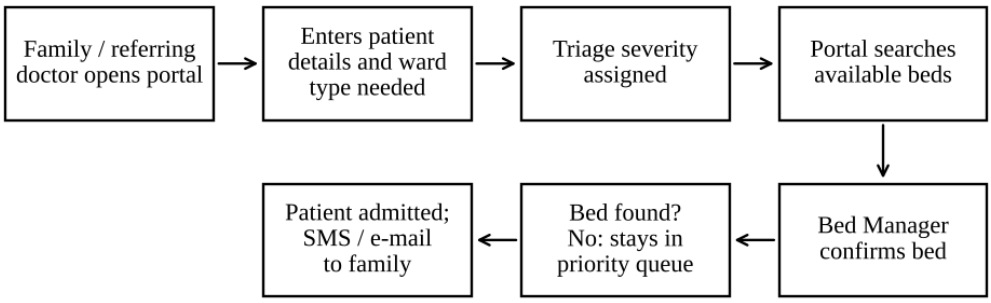


Figure 5: Process map of an online bed request

5.2 E-Payment Integration

Method	How it works	Control
UPI	Family pays advance or final bill using a UPI app or QR code	Bank PIN; transaction reference saved in txn_ref
Debit / Credit card	Card details entered on the gateway page	3-D Secure OTP, card data tokenised
Government health scheme	Eligible patient admitted on scheme card; claim raised by hospital	Scheme authorisation number recorded
Insurance	Cashless claim to the insurer	Pre-authorisation number recorded
Cash at hospital	Counter payment with printed receipt	Receipt number, cashier accountability

Table 6: Payment methods supported

The portal does not store card numbers. The gateway returns only a status and a reference number, which are saved in the payment table.

5.3 Security Architecture

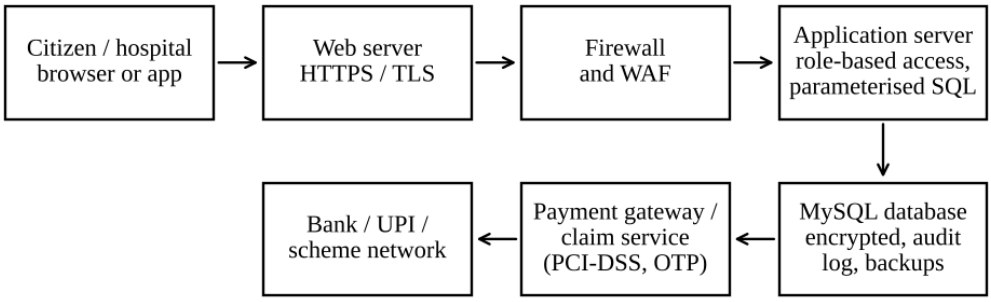


Figure 6: Security architecture of the portal

Security is layered: traffic is encrypted by HTTPS, a firewall filters requests, the application enforces role-based access, queries are parameterised to prevent SQL injection, and sensitive data are encrypted, logged and backed up.

5.4 Technology Adoption

The success of the portal depends on its acceptance by hospitals and staff. The Technology Acceptance Model (TAM) says adoption depends on perceived usefulness and perceived ease of use. Rogers' Diffusion of Innovations classifies adopters into innovators, early adopters, early majority, late majority and laggards.

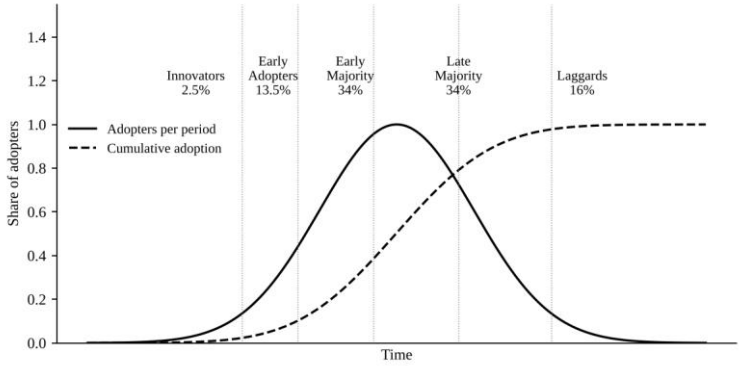


Figure 7: Technology adoption curve (Rogers)

- Innovators and early adopters (government medical college and large private hospitals): pilot the portal and give feedback.
- Early majority (district and taluk hospitals): training, a simple mobile screen for bed status, and a help desk.
- Late majority and laggards (small private hospitals): policy mandate, link to scheme empanelment, and visible benefits such as referral of paying patients.
- Families and doctors: public search page, Kannada and English screens, and a telephone helpline for those without internet access.

6. MODULE 5: IT ETHICS, SECURITY AND AUDIT

6.1 Data Privacy and Ethical Use

Health data is among the most sensitive personal data. The portal holds names, ages, contact details and the ward needed, which can reveal a medical condition. The relevant Indian laws are the Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023 together with its rules. The principal ethical duties are to collect only the minimum data, obtain consent, keep it confidential, and use it only for care and planning. The following controls are applied:

- Consent notice at registration; only the minimum fields needed for allocation are collected.
- Role-based access: hospitals see only their own patients; the control room sees requests; officials see only aggregated reports.
- Patient identity masked in public dashboards; reports use counts, not names.
- Audit log of every allocation and status change to ensure fair, transparent decisions.
- Retention schedule and secure deletion or anonymisation of old records.

Fairness. Allocation is by severity rank and then waiting time, never by wealth or influence. Any override by the control room must be recorded with a reason.

6.2 Cybercrime Risk Matrix

Risk	Likelihood	Impact	Mitigation
Wrong or stale bed status	High	High	Mandatory update at admission and discharge; time-stamp and alert if not updated
SQL injection	Medium	High	Parameterised queries, input validation
Ransomware at a hospital	Medium	High	Offline backups, patching, downtime procedure on paper
Phishing of staff credentials	High	High	Awareness training, two-factor login
Denial of service in a surge	Medium	High	Load balancing, rate limits, helpline backup
Insider snooping on patients	Medium	Medium	Role-based access, audit logs, penalties

Table 7: Cybercrime and operational risk matrix

6.3 Ethical, Ergonomic and Environmental Audit

Area	Observation / Requirement	Action
Ethical use	Health data must stay confidential	Written data-use policy; staff confidentiality undertaking
Accuracy	A wrong bed count can delay care	Update rules, daily reconciliation by bed managers
Ergonomics	Control-room operators work long shifts at screens	Adjustable chairs, monitor at eye level, rotating shifts, 20-20-20 eye rest rule
Lighting and glare	Night shifts cause eye strain	Anti-glare screens, indirect lighting
Environment	Paper forms and registers	Paperless request and discharge records
E-waste	Replaced computers and printers	Authorised e-waste recyclers
Energy	24x7 servers and workstations	Energy-efficient hardware, cloud or shared data centre

Table 8: Ethics, ergonomic and environmental audit

7. MODULE 6: APPLICATION SOFTWARE - MYSQL IMPLEMENTATION

This chapter presents the working database. All code was executed on a MySQL-compatible server and the outputs shown are the actual results. The complete script is submitted as the file icu_bed_portal.sql. All data is fictitious.

7.1 Creating the Database and Tables (DDL)

Input (SQL):

```
CREATE TABLE region (
  region_id   INT AUTO_INCREMENT PRIMARY KEY,
  region_name VARCHAR(40) NOT NULL UNIQUE
);
CREATE TABLE facility_type (
  type_id   INT AUTO_INCREMENT PRIMARY KEY,
  type_name VARCHAR(40) NOT NULL UNIQUE
);
CREATE TABLE facility (
  facility_id   INT AUTO_INCREMENT PRIMARY KEY,
  facility_name VARCHAR(70) NOT NULL UNIQUE,
  type_id       INT NOT NULL,
  region_id     INT NOT NULL,
  ownership     ENUM('Government','Private','Trust') NOT NULL,
  phone         VARCHAR(15),
  FOREIGN KEY (type_id)   REFERENCES facility_type(type_id),
  FOREIGN KEY (region_id) REFERENCES region(region_id)
);
CREATE TABLE ward_type (
  wtype_id   INT AUTO_INCREMENT PRIMARY KEY,
  wtype_name VARCHAR(30) NOT NULL UNIQUE
);
CREATE TABLE facility_ward_rate (
  facility_id  INT NOT NULL,
  wtype_id     INT NOT NULL,
  daily_charge DECIMAL(9,2) NOT NULL CHECK (daily_charge > 0),
  PRIMARY KEY (facility_id, wtype_id),
  FOREIGN KEY (facility_id) REFERENCES facility(facility_id),
  FOREIGN KEY (wtype_id)    REFERENCES ward_type(wtype_id)
);
CREATE TABLE bed (
  bed_id      INT AUTO_INCREMENT PRIMARY KEY,
  facility_id INT NOT NULL,
  wtype_id    INT NOT NULL,
  bed_label   VARCHAR(10) NOT NULL,
  status      ENUM('Available','Occupied','Reserved','Maintenance') DEFAULT 'Available',
  UNIQUE (facility_id, bed_label),
  FOREIGN KEY (facility_id) REFERENCES facility(facility_id),
  FOREIGN KEY (wtype_id)    REFERENCES ward_type(wtype_id)
);
```

Input (SQL):

```
CREATE TABLE severity (
  severity_id   INT AUTO_INCREMENT PRIMARY KEY,
  severity_name VARCHAR(20) NOT NULL UNIQUE,
  priority_rank INT NOT NULL
);
CREATE TABLE patient (
  patient_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name  VARCHAR(60) NOT NULL,
  age        INT NOT NULL CHECK (age >= 0),
  gender     ENUM('M','F','O') NOT NULL,
  phone      VARCHAR(15),
  city       VARCHAR(40)
);
CREATE TABLE staff (
  staff_id    INT AUTO_INCREMENT PRIMARY KEY,
  staff_name  VARCHAR(60) NOT NULL,
  role        VARCHAR(40) NOT NULL,
  facility_id INT NULL,
  FOREIGN KEY (facility_id) REFERENCES facility(facility_id)
);
CREATE TABLE bed_request (
  req_id      INT AUTO_INCREMENT PRIMARY KEY,
  patient_id  INT NOT NULL,
  wtype_id    INT NOT NULL,
  severity_id INT NOT NULL,
  requested_at DATETIME NOT NULL,
  status      ENUM('Pending','Allocated','Rejected','Cancelled') DEFAULT 'Pending',
  staff_id    INT NULL,
  FOREIGN KEY (patient_id)  REFERENCES patient(patient_id),
  FOREIGN KEY (wtype_id)    REFERENCES ward_type(wtype_id),
  FOREIGN KEY (severity_id) REFERENCES severity(severity_id),
  FOREIGN KEY (staff_id)    REFERENCES staff(staff_id)
);
CREATE TABLE admission (
  adm_id       INT AUTO_INCREMENT PRIMARY KEY,
  req_id       INT NOT NULL UNIQUE,
  bed_id       INT NOT NULL,
  admitted_at  DATETIME NOT NULL,
  discharged_at DATETIME NULL,
  outcome      ENUM('Recovered','Referred','Deceased') NULL,
  FOREIGN KEY (req_id) REFERENCES bed_request(req_id),
  FOREIGN KEY (bed_id) REFERENCES bed(bed_id)
);
CREATE TABLE payment (
  payment_id INT AUTO_INCREMENT PRIMARY KEY,
  adm_id     INT NOT NULL,
  amount     DECIMAL(10,2) NOT NULL CHECK (amount > 0),
  method     ENUM('Cash','UPI','Insurance','Govt Scheme') NOT NULL,
```

```
  pay_date   DATE NOT NULL,
  txn_ref    VARCHAR(30),
  FOREIGN KEY (adm_id) REFERENCES admission(adm_id)
);
```

7.2 Inserting Data (Input)

Input (SQL):

```
INSERT INTO region (region_name) VALUES
 ('Bengaluru Urban'),('Bengaluru Rural'),('Mysuru'),('Tumakuru'),('Mandya');
INSERT INTO ward_type (wtype_name) VALUES
 ('General ICU'),('ICU with Ventilator'),('NICU'),('PICU'),('Cardiac ICU');
INSERT INTO facility (facility_name, type_id, region_id, ownership, phone) VALUES
 ('Cauvery District Hospital',2,3,'Government','0821-2400001'), ...;
```

Facility types, severity levels, staff, facility ward rates and patients were inserted in the same way. Beds, requests and admissions are shown below in abbreviated form; the full lists are in the submitted .sql file.

Input (SQL):

```
INSERT INTO bed (facility_id, wtype_id, bed_label, status) VALUES
 (1,1,'A1','Occupied'),(1,1,'A2','Available'),
 (1,2,'V1','Occupied'),(1,2,'V2','Available'), ...;
INSERT INTO bed_request (patient_id, wtype_id, severity_id, requested_at, status,
staff_id) VALUES
 (1,1,2,'2026-10-20 08:10','Allocated',2), ...,
 (18,2,1,'2026-11-05 08:40','Pending',NULL), ...;
```

7.3 Bill View

A view named v_bill calculates, for every admission, the number of days (minimum one), the total bill from the facility's daily charge for that ward type, and the amount paid. It is used by the billing and DSS queries.

Input (SQL):

```
CREATE VIEW v_bill AS
SELECT a.adm_id, p.full_name, f.facility_name, wt.wtype_name,
       GREATEST(1, DATEDIFF(COALESCE(a.discharged_at,'2026-11-05 10:00:00'),
       a.admitted_at)) AS days,
       GREATEST(1, DATEDIFF(COALESCE(a.discharged_at,'2026-11-05 10:00:00'),
       a.admitted_at)) * fr.daily_charge AS total_bill,
       COALESCE(pm.paid, 0) AS paid
FROM admission a
JOIN bed_request r ON r.req_id = a.req_id
JOIN patient p     ON p.patient_id = r.patient_id
JOIN bed b         ON b.bed_id = a.bed_id
JOIN facility f    ON f.facility_id = b.facility_id
JOIN ward_type wt  ON wt.wtype_id = b.wtype_id
JOIN facility_ward_rate fr ON fr.facility_id = b.facility_id AND fr.wtype_id = b.wtype_id
```

```
LEFT JOIN (SELECT adm_id, SUM(amount) AS paid FROM payment GROUP BY
adm_id) pm
      ON pm.adm_id = a.adm_id;
```

Input (SQL):

```
INSERT INTO payment (adm_id, amount, method, pay_date, txn_ref) VALUES
 (1,1000,'UPI','2026-10-20','UPI66010001'), ...;
INSERT INTO payment (adm_id, amount, method, pay_date, txn_ref)
SELECT b.adm_id, b.total_bill - b.paid, 'Govt Scheme', DATE(a.discharged_at),
  CONCAT('SETTLE-', b.adm_id) -- full script picks method by ownership
FROM v_bill b JOIN admission a ON a.adm_id = b.adm_id
WHERE a.discharged_at IS NOT NULL AND b.total_bill > b.paid;
```

7.4 Portal Operations (Input and Output)

Operation 1 - Search available ICU-with-ventilator beds across all regions.

Input (SQL):

```
SELECT f.facility_name, r.region_name, b.bed_label, fr.daily_charge
FROM bed b
JOIN facility f ON f.facility_id = b.facility_id
JOIN region r    ON r.region_id = f.region_id
JOIN facility_ward_rate fr ON fr.facility_id = b.facility_id AND fr.wtype_id = b.wtype_id
WHERE b.wtype_id = 2 AND b.status = 'Available'
ORDER BY fr.daily_charge;
```

Output:

facility_name	region_name	bed_label	daily_charge
Namma Medical College Hospital	Bengaluru Urban	V1	2500.00
Cauvery District Hospital	Mysuru	V2	2500.00
Gateway Super-speciality Hospital	Bengaluru Rural	V1	12500.00

Operation 2 - Register a new bed request (new patient).

Input (SQL):

```
INSERT INTO patient (full_name, age, gender, phone, city)
VALUES ('Mohan Das', 59, 'M', '9845012021', 'Bengaluru');
INSERT INTO bed_request (patient_id, wtype_id, severity_id, requested_at, staff_id)
VALUES (LAST_INSERT_ID(), 1, 2, '2026-11-05 10:05:00', 2);
```

Output:

req_id	full_name	wtype_name	severity_name	status
23	Mohan Das	General ICU	Serious	Pending

Operation 3 - Pending requests in priority order (severity, then waiting time).

Input (SQL):

```
SELECT r.req_id, p.full_name, wt.wtype_name, s.severity_name,
       TIMESTAMPDIFF(MINUTE, r.requested_at, '2026-11-05 10:10:00') AS waiting_min
FROM bed_request r
JOIN patient p    ON p.patient_id = r.patient_id
JOIN ward_type wt ON wt.wtype_id = r.wtype_id
JOIN severity s   ON s.severity_id = r.severity_id
WHERE r.status = 'Pending'
ORDER BY s.priority_rank, r.requested_at;
```

Output:

req_id	full_name	wtype_name	severity_name	waiting_min
18	Rajesh Naik	ICU with Ventilator	Critical	90
20	Baby of Anjali	NICU	Critical	40
19	Meena Kumari	General ICU	Serious	60
23	Mohan Das	General ICU	Serious	5

Operation 4 - Allocate a bed to the first request (one transaction updates three tables).

Input (SQL):

```
START TRANSACTION;
INSERT INTO admission (req_id, bed_id, admitted_at) VALUES (18, 22, '2026-11-05 10:20:00');
UPDATE bed SET status = 'Occupied' WHERE bed_id = 22;
UPDATE bed_request SET status = 'Allocated', staff_id = 2 WHERE req_id = 18;
COMMIT;
```

Output:

req_id	req_status	bed_id	bed_status	facility_name
18	Allocated	22	Occupied	Gateway Super-speciality Hospital

Operation 5 - Check the bill of a patient and discharge with final payment.

Input (SQL):

```
SELECT full_name, wtype_name, days, total_bill, paid, total_bill - paid AS balance
FROM v_bill WHERE adm_id = 7;
```

Output:

full_name	wtype_name	days	total_bill	paid	balance
Kamala Bai	General ICU	5	7500.00	1500.00	6000.00

Input (SQL):

```
START TRANSACTION;
UPDATE admission SET discharged_at = '2026-11-05 11:00:00', outcome = 'Recovered'
WHERE adm_id = 7;
INSERT INTO payment (adm_id, amount, method, pay_date, txn_ref)
SELECT adm_id, total_bill - paid, 'Govt Scheme', '2026-11-05', 'AB-ARK-0107'
FROM v_bill WHERE adm_id = 7;
UPDATE bed SET status = 'Available' WHERE bed_id = 1;
COMMIT;
```

Output:

adm_id	outcome	bed_status	balance_due
7	Recovered	Available	0.00

After discharge the balance due is zero and the bed status returns to Available, so the next request in the queue can be matched to it immediately.

7.5 Decision Support (DSS) Queries

Query 1 - Region-wise bed availability and occupancy (JOIN, GROUP BY).

Input (SQL):

```
SELECT rg.region_name, COUNT(*) AS total_beds,
       SUM(b.status = 'Available') AS available,
       SUM(b.status = 'Occupied')  AS occupied,
       ROUND(100 * SUM(b.status = 'Occupied') / COUNT(*), 1) AS occupancy_pct
FROM bed b
JOIN facility f ON f.facility_id = b.facility_id
JOIN region rg  ON rg.region_id = f.region_id
GROUP BY rg.region_name
ORDER BY occupancy_pct DESC;
```

Output:

region_name	total_beds	available	occupied	occupancy_pct
Bengaluru Urban	12	5	6	50.0
Bengaluru Rural	4	2	2	50.0
Mandya	2	1	1	50.0
Mysuru	7	4	2	28.6
Tumakuru	2	1	0	0.0

Query 2 - Availability by ward type.

Input (SQL):

```
SELECT wt.wtype_name, COUNT(*) AS total_beds,
       SUM(b.status = 'Available') AS available,
       SUM(b.status = 'Occupied')  AS occupied
FROM bed b JOIN ward_type wt ON wt.wtype_id = b.wtype_id
```

```
GROUP BY wt.wtype_name
ORDER BY available;
```

Output:

wtype_name	total_beds	available	occupied
PICU	2	1	1
NICU	3	1	1
Cardiac ICU	4	2	2
ICU with Ventilator	6	2	4
General ICU	12	7	3

Query 3 - Facilities with occupancy of 50 per cent or more (GROUP BY with HAVING).

Input (SQL):

```
SELECT f.facility_name, f.ownership, COUNT(*) AS beds,
       ROUND(100 * SUM(b.status = 'Occupied') / COUNT(*), 1) AS occupancy_pct
FROM bed b JOIN facility f ON f.facility_id = b.facility_id
GROUP BY f.facility_id, f.facility_name, f.ownership
HAVING occupancy_pct >= 50
ORDER BY occupancy_pct DESC;
```

Output:

facility_name	ownership	beds	occupancy_pct
Namma Medical College Hospital	Government	5	60.0
Shivanasamudra Taluk Hospital	Government	2	50.0
Sunrise Multi-speciality Hospital	Private	4	50.0
Gateway Super-speciality Hospital	Private	4	50.0

Query 4 - Collections by ownership of facility.

Input (SQL):

```
SELECT f.ownership, COUNT(DISTINCT a.adm_id) AS admissions,
       SUM(p.amount) AS amount_collected
FROM payment p
JOIN admission a ON a.adm_id = p.adm_id
JOIN bed b       ON b.bed_id = a.bed_id
JOIN facility f  ON f.facility_id = b.facility_id
GROUP BY f.ownership
ORDER BY amount_collected DESC;
```

Output:

ownership	admissions	amount_collected
Private	6	171000.00
Government	7	46900.00

Query 5 - Patients whose bill is above the average bill (sub-query).

Input (SQL):

```
SELECT full_name, facility_name, total_bill
FROM v_bill
WHERE total_bill > (SELECT AVG(total_bill) FROM v_bill)
ORDER BY total_bill DESC;
```

Output:

full_name	facility_name	total_bill
Baby of Radha K	Gateway Super-speciality Hospital	72000.00
Mahesh Gowda	Sunrise Multi-speciality Hospital	44000.00
Gopal Rao	Sunrise Multi-speciality Hospital	24000.00
Venkatesh Prasad	Kaveri Heart and Critical Care	23000.00
Lakshmi Devi	Sunrise Multi-speciality Hospital	22000.00

Query 6 - Executive summary (scalar sub-queries).

Input (SQL):

```
SELECT (SELECT COUNT(*) FROM bed)                          AS total_beds,
       (SELECT COUNT(*) FROM bed WHERE status='Available')   AS available,
       (SELECT COUNT(*) FROM bed WHERE status='Occupied')    AS occupied,
       (SELECT COUNT(*) FROM bed_request WHERE status='Pending') AS
       pending_requests,
       (SELECT ROUND(AVG(total_bill),0) FROM v_bill)        AS avg_bill;
```

Output:

total_beds	available	occupied	pending_requests	avg_bill
27	13	11	3	15411

7.6 Executive Report - Interpretation

- **Capacity:** Of 27 critical-care beds, 13 are available, 11 occupied, 2 reserved and 1 under maintenance. Bengaluru Urban, Bengaluru Rural and Mandya are at 50 per cent occupancy, while Mysuru is at 28.6 per cent, so patients from crowded regions could be referred there.
- **Scarcest resource:** Only 2 of 6 ICU-with-ventilator beds are free, the tightest ward type, whereas 7 of 12 general ICU beds are free. Investment should target ventilator beds.

- **Cost:** Private facilities collected Rs. 1,71,000 for 6 admissions against Rs. 46,900 for 7 admissions in government facilities, showing the importance of scheme support for affordable care.

Note: figures are based on a small sample for demonstration and should not be taken as real hospital statistics.

8. FINDINGS, CONCLUSION AND FUTURE SCOPE

8.1 Findings

- A normalised database avoided repetition of hospital, region, ward and rate data and prevented inconsistent records.
- SQL aggregate queries gave an instant view of regional capacity and the shortage of ventilator beds that earlier required many phone calls.
- Severity-based ranking makes allocation transparent and auditable.

8.2 Conclusion

The Regional Healthcare Facilities and ICU Bed Portal demonstrates all six modules of Technology for Management: classification of the system as TPS, MIS and DSS; a system view and 3NF relational design; SDLC, BPR and data-flow modelling; e-governance, payment security and adoption planning; privacy, ethical and ergonomic considerations; and a working MySQL application. A shared, accurate and real-time bed database can reduce the time families spend searching for care and help the government plan critical-care capacity.

8.3 Limitations and Future Scope

- The model uses a fixed daily charge and sample data from eight facilities.
- Future work includes a mobile app, integration with ambulance dispatch and hospital information systems, distance-based matching using maps, and AI-based demand forecasting.

9. REFERENCES AND INDIVIDUAL ATTRIBUTION LOG

9.1 Individual Attribution Log

As per the project guidelines, each student owns a module matrix. Replace the placeholders with the names of the group members.

Student	Modules Owned	Contribution
Student 1 (Data and RDBMS Engineer)	2 and 6	ER diagram, 3NF schema, MySQL scripts, DSS queries
Student 2 (Systems Analyst)	1 and 3	System classification, SDLC, BPR, context diagram, DFD
Student 3 (E-Governance and Ethics Specialist)	4 and 5	Payment and security framework, adoption analysis, ethics audit, report consolidation

Table 9: Individual attribution log (3-member group)

9.2 References

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